

1(a)(i)	B and C ; Accept A, B and C <i>if any other letters do not award the mark, ignore names</i>	1
1(a)(ii)	any two from: keeps the (named) airway(s) open / prevents collapse of airways ; AW R bronchiole / alveoli e.g. to allow free flow of air into and out of, airways / lungs e.g. provides support (for the named airway(s)) I 'provides strength' allows flexibility ; A described e.g. bending neck, swallowing food AVP ; e.g. (C-shaped incomplete) rings (in, B / trachea) allow, lengthening / widening, during breathing in / inspiration / inhalation	2

Question	Answer				Marks
1(a)(iii)	<i>region</i> <i>one mark for each letter – for 1.3 allow two or three letters if all are correct – ignore names</i> <i>one mark for feature matched with function – feature and relationship to function must match</i> <i>if find a correct feature and its function in the last column – accept</i> <i>feature must be visible in Fig. 1.2 and Fig. 1.3</i> <i>if gas (as in gas exchange) is not used, then answer must refer to oxygen and carbon dioxide</i>				4
	Fig.	region of the gas exchange system (A, B, C, D, E, F, G)	one visible feature	one way in which the feature relates to its function	
	1.2	G ;	air spaces walls are, thin / one cell thick A squamous epithelium R cell wall good blood supply / capillaries A capillary / red blood cells arteriole / venule R artery / vein	large surface area for, diffusion / gas exchange ; short distance for, diffusion / gas exchange ; for, diffusion / gas exchange or maintains steep concentration gradients or absorption / transport, of oxygen ; supply blood to capillaries / carry blood away from capillaries ;	
	1.3	B ; Accept C / A	goblet cells (many) cilia / ciliated epithelial cells / ciliated epithelium / ciliated cells ciliated epithelium	secrete / produce / make, mucus ; (cilia) move / AW, mucus (towards back of throat / AW) ; provides barrier to pathogens ;	

Question	Answer	Marks
1(b)	any five from: 1 sinoatrial node, releases / generates / produces, impulses / waves of excitation / waves of depolarisation / AW ; I 'signals' A SAN / SA node / pacemaker R nervous impulses 2 impulses / AW, spreads to / across, walls / muscle of, (right) atrium ; A atria 3 <u>right atrium</u> contracts ; A atrial systole if right atrium is given in answer, A pumps for contracts 4 tricuspid / (right) atrioventricular, valve <u>opens</u> to allow blood to flow (from right atrium) into <u>right ventricle</u> ; R bicuspid valve / left atrioventricular valve 5 atrioventricular node sends impulses down, septum / Purkyne tissue ; A to apex of heart / down Bundle of His A AVN / AV node 6 atrioventricular node is responsible for / AW, a time delay ; A AVN / AV node 7 <u>right ventricle</u> contracts ; A ventricular systole if right ventricle is given in answer, A pumps for contracts 8 semi-lunar valve <u>opens</u> and blood flows into <u>pulmonary</u> artery / arteries ; A pulmonary valve R aorta 9 AVP ; e.g. tricuspid / (right) atrioventricular, valve <u>closes</u> to stop, backflow / blood entering right atrium (only in correct context)	5